

ECOMIX LLP · EASYMIX · KARATOBE FIELD, J1-IV HORIZON

NANOCEM UT-9

An ultra-fine injection grout for water shut-off. A D95 of 9 µm or finer lets the compound enter channels that ordinary oil-well cement cannot reach.

PARTICLE FINENESS

D95 ≤ 9 µm

D50 ~3.5 µm

28-DAY STRENGTH

65–75 MPa

grade M500

TREATMENTS AT KARATOBE

7 wells

January — September 2026

REFERENCE RESULT

1.1 → 16.4

t/day on well 343

THE WELL PRODUCES WATER, NOT OIL

On a mature field water cut reaches 80–95%. The pump lifts liquid, but only a few percent of it is oil: the cost of lifting, treating and disposing of water rises while production falls.

Water reaches the bottomhole through microfractures and fine filtration channels — a matter of single and tens of microns. Ordinary oil-well cement cannot cut them off: its particles are larger than the channels themselves.

FROM THE FIELD

Well 343, January 2026

1.10 t/day at 89.4% water cut. The well was on the verge of being shut in — liquid was being lifted, but almost no saleable oil.

FROM THE FIELD

Well 357, June 2026

1.50 t/day at 80.6% water cut. Logging confirmed it directly: of 86.4 m³ of displaced liquid, 72.7 m³ was formation water.

WHAT IS NEEDED

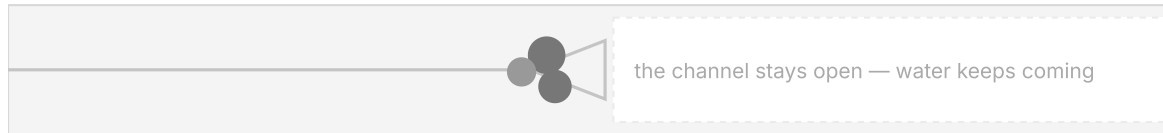
Selective isolation

Close the watered-out intervals without killing the oil-bearing part of the reservoir. The well has to come back on stream, not stop.

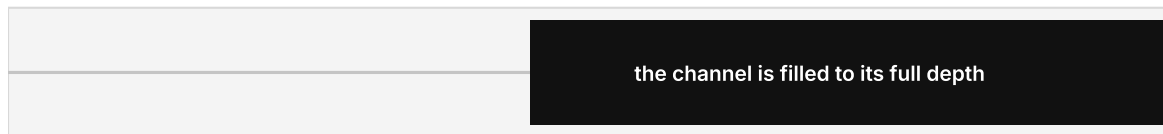
FINENESS DECIDES IT, NOT PUMP PRESSURE

The difference between “the grout entered the fracture” and “the grout bridged off at its mouth” is particle size.

ORDINARY CEMENT · 40–80 μm



NANOCEM UT-9 · $D_{95} \leq 9 \mu\text{m}$



Ordinary cement particles of 40–80 μm bridge off at the mouth of the channel without filling it: deeper in, the water path stays open. In NanoCem UT-9 the overwhelming majority of particles are finer than 9 μm — the grout travels in and fills the channel completely. Once set, it is not a plug but structural stone.

PRODUCT DATA SHEET

TU 5745-001-171140033124-2025 · dry blend, grey powder · 25 kg / 1000 kg

Fineness D50 / D95	~3.5 µm / ≤ 9 µm
Dry product density	2.95 g/cm³
Slurry density	1.85–1.95 g/cm³
Working life of the slurry	4–5 hours
Initial set	~60 min
Full set	24 hours at +25 °C
Strength after 24 hours	≥ 12 MPa

Strength after 28 days	65–75 MPa · grade M500
Bond to concrete	≥ 2.5 MPa
Water resistance	W14–W16
Layer thickness	5–100 mm
Chemical resistance	oil, brines, sulphates
Shelf life	12 months

WHAT THE COMPOUND SEALS

Three uses proven on the Karatobe well stock.

USE 01

Water shut-off

Cutting off watered-out intervals and the channels feeding water to the bottomhole. The oil-bearing part of the reservoir keeps working — the well returns to production.

USE 02

Crossflow behind casing

Sealing the channels behind the casing through which water travels from horizons above or below. The ultra-fine grind fills narrow gaps.

USE 03

Repairing the cement sheath

Filling cracks and micro-gaps in the sheath and restoring zonal isolation. A bond of 2.5 MPa or better grips the existing stone.

HOW IT IS DONE AT THE WELL

The established squeeze sequence — as run on well 357, September 2026

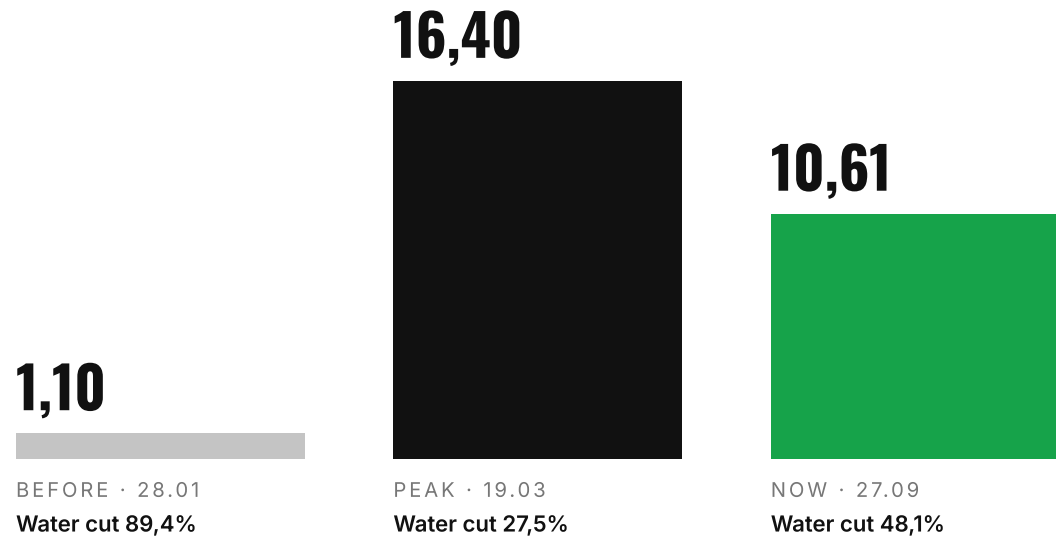
01 Preparation Shut in, log, identify the inflow intervals. The slurry is mixed at a water-to-cement ratio of about 0.87 — deliberately mobile, or it will not enter the channels it is chosen for.	02 A three-stage squeeze First a portion with the annulus open — the grout is distributed across all the intervals. Then the tubing is pulled and a second portion goes in. Only then is the packer set and the rest squeezed in closed.	03 Pressure as the control On the closed-in stage the pressure builds and then stops: the formation has stopped taking. That is the working sign of a seal — while the channels are open, pressure does not build at a constant pump rate.
04 Waiting on cement under pressure The well stays under pressure throughout — the compound gains strength under compression rather than simply standing in the hole.	05 Drill-out and pressure test The bridge is drilled out and each interval is tested separately as it is opened, so a leaking interval is located immediately.	06 Perforation and start-up Logging, perforation, ramp-up. The effect is judged from the water-cut curve over two to three weeks.

SEVEN TREATMENTS AT KARATOBE

J1-IV horizon · operator IC Petroleum · daily operating logs as of 27.09.2026

WELL	ENTRIES	ISOLATION OUTCOME	DATE	RESULT
343	1	● Tight on the 1st	Jan 2026	1.10 → 16.40 t/day at the peak · water cut 89.4 → 27.5%
301	1	● Tight on the 1st	06.06.2026	10.28 t/day · water cut 37.1%
303	1	● Tight on the 1st	13.07.2026	2.86 t/day — the well's record · water cut 95 → 65%
342	2	● Tight on the 2nd	10.05 → 31.05	7.86 t/day · water cut 37.1% · ×13.3 over baseline
311	1	● Confirmed after ramp-up	09.07.2026	water cut 90.5 → 49%, settled at 60%
357	1	● Tight on pressure test	24.09.2026	direct test: 3 of 3 intervals hold
305	1	● Cannot be assessed — target changed	18.07.2026	switched to the T1-II horizon in September

WELL 343



Treated in January 2026. The rate rose almost fifteenfold and water cut fell from 89.4% to 27.5% at the peak. The effect is holding into its eighth month.

Oil rate, t/day. Eighth month after the shut-off.

PRODUCED SINCE THE JOB

≈**3,033 t**

over 235 days on stream

GAIN OVER BASELINE

×**9.6**

1.10 → 10.61 t/day

WATER CUT

89.4 → 48.1%

27.5% at the peak

EFFECT DURATION

8 months

and continuing

AN HONEST LIMIT OF APPLICABILITY

Isolating the perforations does not help if the water travels past them — behind the casing.

Logging reports on two edge wells show poor cement behind casing: continuous contact over only 6.25% and 30.4% of the hole. With contact like that, part of the water moves behind the casing along the hole, bypassing the perforations, and closing the holes does not produce a seal.

INDICATOR

Poor cement behind casing

Continuous contact over less than a third of the hole on the bond log. The sheath must be assessed before the job, or the result will be partial.

INDICATOR

An edge position in the pool

The central-eastern core seals on the first entry. At the edges a second entry or a different scheme is required.

WHAT FOLLOWS

Candidate selection

Water cut on its own is a poor predictor. Candidates should be picked by position and casing condition, not by the water percentage.

TIGHTNESS CAN BE PROVEN STRAIGHT AWAY

On well 357 the isolation was confirmed by direct test — before start-up, without waiting for production statistics.

Cement bridge	40 atm · 15 min	tight
Intervals 657–663 and 669.5–672 m	30 atm	tight
Interval 679–684 m	40 atm · 15 min	tight

The testing was done in stages as the bridge was drilled out: each interval was checked as soon as it was opened. Testing the whole hole at once would not locate a leaking interval. A caveat: the test was at 30–40 atm against a squeeze pressure of 80 atm, and it says nothing about crossflow behind the casing.

WHAT WE OFFER

The compound, a proven squeeze procedure and transparent analytics on every treatment.

01

The compound

An ultra-fine blend of our own manufacture, TU 5745-001-171140033124-2025. Supplied in 25 kg and 1000 kg packs, 12-month shelf life.

02

The procedure

A three-stage squeeze with pressure control, a separate pressure test on each interval and subsequent re-perforation.

03

The analytics

Daily monitoring of the well stock, effect measured from the water-cut curve, failures analysed. All of it open at nanocem.app.

NANOCEM.APP — THE WELL STOCK, THE TIGHTNESS REVIEW AND THE WORK PROGRAMME